

Thank you for purchasing a FANUC Robot.

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation provided with your robot for safety procedures.

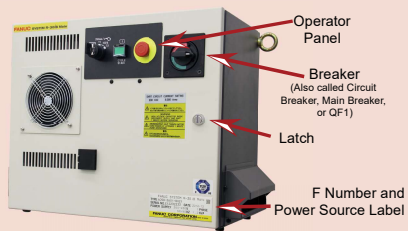
Required Tools

- ◆ Straight-head screwdriver ◆ Torque wrench
- ◆ Cross-head screwdriver ◆ Metric hex-head wrenches

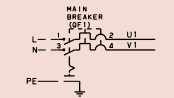
Required Items

- ◆ Properly rated transport equipment as specified
- ◆ Customer-supplied properly rated power source as specified on the Controller Power Source Label
- ◆ Customer-supplied properly rated input power cable; conductor size of AWG14 to AWG10; terminal size is M5
- ◆ Multimeter ◆ Rigid platform to hold robot
- ◆ 4-M10 x 35mm bolts ◆ Loctite#243
- ◆ 4-M10 flat washers

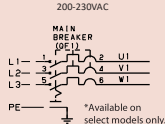
Mate Controller (Open Air controller not shown)



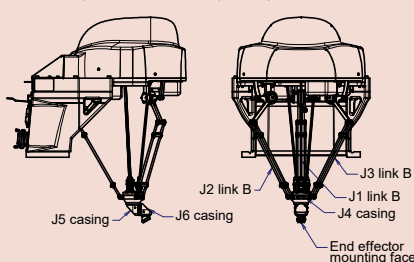
Single Phase



Three Phase*



Robot (mechanical unit) with joints indicated



M-1iA/0.5A 6-Axis Robot shown
(other models might have less than 6 axes)

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A. Unpack and Install your Controller

Shipment Contents



1. Squeeze and pull the clips off of the skid and remove the cardboard cover. Open all of the boxes included in your shipment. Unwrap the robot.

Option Box (optional)



Accessories Box



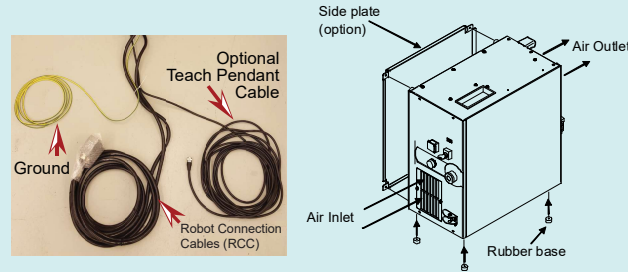
Hardware Bag, Teach Pendant (if ordered), & Inspection Data Sheet.

Store these items in a safe, known location.

3. Place the controller in a safe location where the grounding wire and RCC can reach the robot, and the controller is safely reachable from outside of the robot work area.

NOTE: Mate Open Air controllers can be installed horizontally or vertically. Ensure there is space for ventilation in the front and rear of the controller.

Vertical Installation

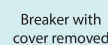
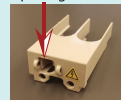
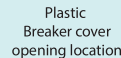


- 4c. Route the user-supplied input power cable through the side of the controller.

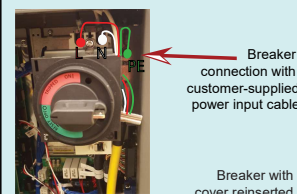


R-30iB Mate Controller - Inside View

- 4d. To connect power, remove the plastic cover from above the Breaker (see the diagram below.) To remove the cover, insert a straight-head screwdriver into the opening, and push the tab to the left allowing the cover to slide forward.



- 4e. Connect power to the Breaker with the customer-supplied input power cable. For Single Phase, connect L to 1 and N to 3. For Three Phase, connect L1 to 1, L2 to 3, and L3 to 5. Connect the ground to the plate located to the right of the Breaker.



- 4f. After the power cable is connected, reinsert the plastic cover.

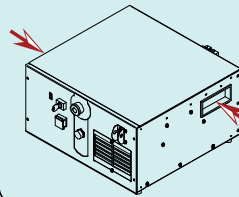
- 4g. Use a multimeter to verify the correct voltage is present.



2. Your shipment includes either the standard Mate controller or the Mate Open Air controller. Carefully remove the packaging to identify the controller for removal.

Mate Open Air Controller

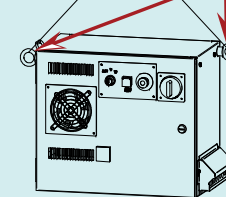
Use the handles on each side to lift up and remove the controller.



Mate Controller
Use a properly rated sling and crane to transport the controller.

Crane capacity: >150kg
Sling capacity: > 150kg

Eyebolt (M10)

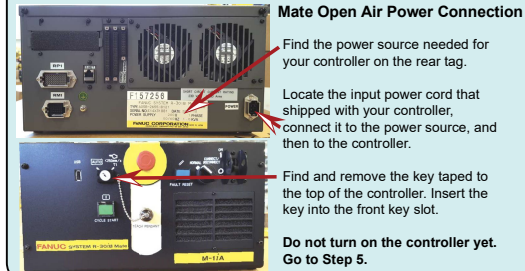


WARNING
Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

4. Follow the controller specific instructions to connect power to the controller.

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.



Mate Open Air Power Connection

Find the power source needed for your controller on the rear tag.

Locate the input power cord that shipped with your controller, connect it to the power source, and then to the controller.

Find and remove the key taped to the top of the controller. Insert the key into the front key slot.

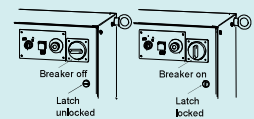
Do not turn on the controller yet. Go to Step 5.

Mate Power Connection

- 4a. Find the label on the bottom right corner of the controller door. Confirm the power source needed for your controller. If a 120VAC step transformer is used, connect it to a 120VAC 20 amp circuit.



- 4b. Open the controller door by using a straight-head screwdriver to turn the latch counter-clockwise, then turn the Breaker counter-clockwise past its off position.



B. Unpack and Install your Robot

WARNING

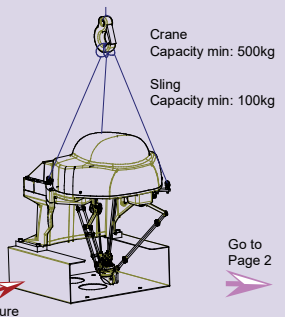
Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

When transporting or installing the robot, be careful not to touch the drive shaft, the links, and the wrist.

Do not use the fall protection fixture to install the robot.

5. Remove the bolts holding the fall protection fixture to the skid.

6. Use a properly rated crane and sling connected to the 3 eye bolts to gently move the robot to its new location.

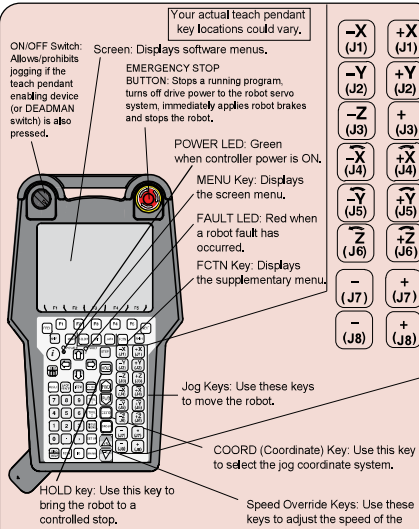


Fall protection fixture

MARGTM1A06171E REV B

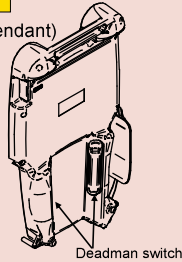
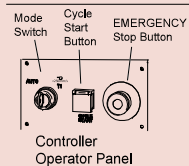
Go to Page 2

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Refer to your Application-Specific Setup and Operations Manual for more information on each teach pendant key.

iPendant (Teach Pendant)



For more information

To access controller, robot, or application-specific manuals, scan the QR Code to the right or go to <https://www.fanucamerica.com/>.

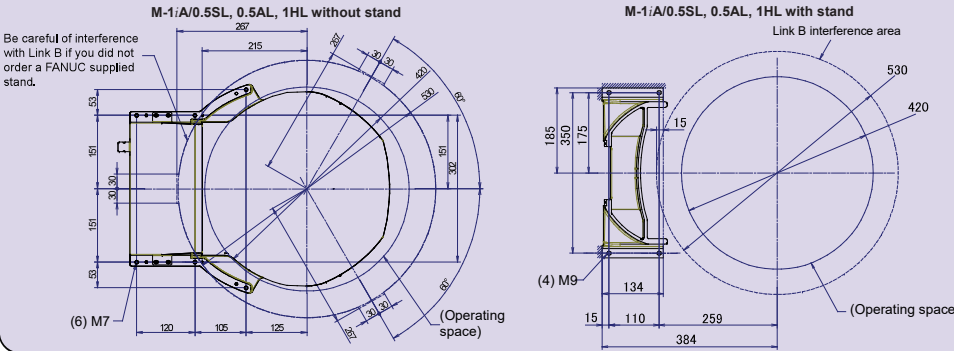


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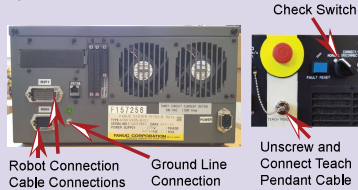
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B. Unpack and Install your Robot continued

7. If using the stand or ceiling mount, use (4) M9 bolts to mount the robot. If no stand is being used, mount the robot with (6) M7 bolts. See the diagram below for the two different patterns. Refer to the robot model-specific Operator's Manual for force data and more information.



9. If you have the Open Air controller, connect the Robot Connection Cable (RCC) and ground line to the controller. If you have a teach pendant, unscrew the silver cover on the front and connect the teach pendant cable. Ensure the CONNECT/DISCONNECT switch is on NORMAL.



Robot installation is complete. You are now ready to go to C. Turn on your Controller.

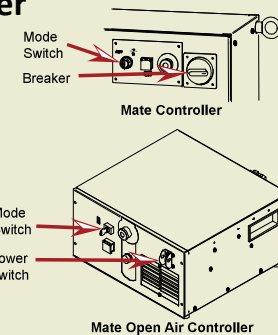
C. Turn on your Controller

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

10. Turn on the Mate controller by rotating the Breaker clockwise to ON. For the Mate Open Air controller, flip the power switch on.

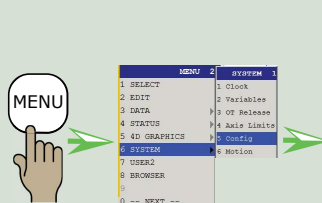
11. Locate the Mode switch on the controller Operator Panel. Use the key retrieved in Step 4 to switch to T1 mode. T1 mode limits the robot speed to 250mm/sec and requires robot control to be at the teach pendant only.



D. Jog your Robot

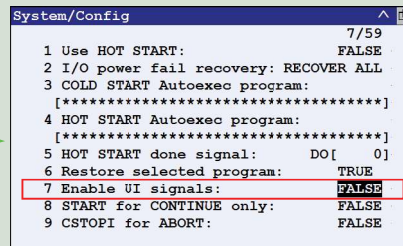
14. To jog the robot in JOINT mode:

a. Turn the Teach Pendant ON/OFF switch to the ON position. Ensure the Emergency Stop buttons on the teach pendant and controller are popped out. If not, twist the button clockwise to reset.



b. Press MENU, select 0 (Next), select (SYSTEM), and then select CONFIG.

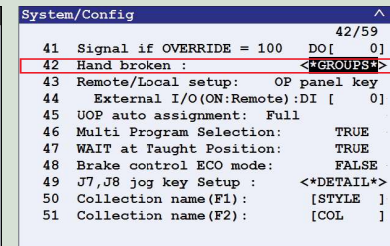
- Use the teach pendant arrow keys to move the cursor to the Enable UI signals option.
- Use the teach pendant F5 key to set the option to FALSE.



c. Use the teach pendant arrow keys to move the cursor to the Hand Broken option (around line 42) and press ENTER.

Hint: You can hold the SHIFT key and press the down arrow key to cursor through the list faster.

- Use the teach pendant F5 key to disable the hand broken error for each motion group.



d. Press and hold a DEADMAN switch on the back of the teach pendant to its middle position. Press RESET on the teach pendant to reset any existing alarms.

NOTE: To view alarms, press MENU, cursor to ALARM, and press ENTER. To toggle between alarm history and active alarms, press F3.

e. To move each joint axis of the robot (called "jogging the robot in JOINT mode"), press and hold SHIFT and press each jog key one at a time and watch the robot joint move. Refer to the Robot (mechanical unit) with joints indicated diagram on Page 1 of this guide.

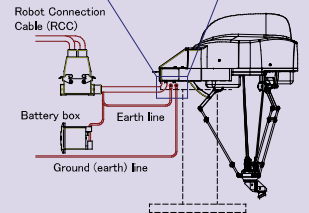
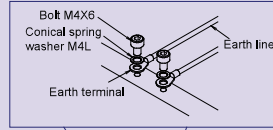
Try jogging each joint of the robot.

Turning on and Jogging the robot is complete. You are now ready to perform software installation, if necessary. Refer to the Controller-specific Software Installation manual.

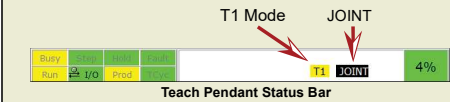
8. Connect the Ground (earth) line and Robot Connection Cable to the robot. If you have a teach pendant, connect it to the teach pendant cable.



NOTE: If the robot was ordered without a stand, connect as shown below.



12. Verify the controller is in T1 Mode by looking at the teach pendant screen. The mode indicator will be highlighted yellow and look similar to the status bar below. If it is not T1, repeat the previous step.



13. Press the COORD key on the teach pendant until JOINT is displayed similar to the status bar above.

You have turned on your controller. You are now ready to go to D. Jog your Robot.